PROJECT PROPOSAL

NUTRIENT DEFICIENCY & THE HIDDEN HUNGER PARADOX IN INDIA

Why High Caloric Intake Still Leads to Severe Micronutrient and Macronutrient Deficiencies Due to Lack of Nutritional Awareness

Team Member Name	Roll Number	Project Role
Yeshwanth	[Your Roll Number 1]	Data Architecture, ETL & Geospatial Modeling
[Team Member 2 Name]	[Roll Number 2]	Statistical Inference & Python Visualizations
[Team Member 3 Name]	[Roll Number 3]	Power BI Model, DAX & Frontend Dashboard

1. Background & Problem Statement

In modern India, food security programs and economic growth have largely satisfied aggregate daily caloric energy requirements (averaging 99.6% to 102.4% fulfillment). However, the nutritional quality of diets remains critically impaired: 74.6% of the population suffers from severe protein deficiency, 56.6% from anemia, and over 75% from vitamin D and B12 deficits. This project empirically investigates the "Hidden Hunger Paradox" to identify how lack of dietary awareness drives overconsumption of empty refined carbohydrates and how localized dietary substitutions can resolve this crisis.

2. Project Objectives

- Compile, clean, and standardize an all-India nutritional fact and dimension dataset across all 36 States & UTs.
- Benchmark empirical nutritional intake against the latest ICMR-NIN Dietary Guidelines for Indians (2024), NFHS-5, and CNNS surveys.
- Engineer clinical diagnostic metrics including the Protein Deficit %, Caloric Sufficiency %, and Hidden Hunger Severity Index.
- Implement a production-grade Star Schema data model for Microsoft Power BI Desktop ingestion.
- Deploy a live interactive web analytics application featuring a 5-tier India Choropleth map, multi-nutrient matrix heatmap, and a personalized body-needs plate calculator with budget-friendly Indian food swaps.

3. Technical Methodology & Tools

- **Data Preprocessing & Cleaning:** Python (Pandas, NumPy, OpenPyXL).
- **Statistical Analysis:** Python (Matplotlib, Seaborn, SciPy).
- **Business Intelligence:** Microsoft Power BI Desktop (Star Schema with 12 DAX measures).
- **Web Application:** HTML5, CSS3, JavaScript (Leaflet.js, Chart.js) deployed live on GitHub Pages.

4. Repository & Live Application Links

Public GitHub Repository: https://github.com/yeshwanthkushi25-glitch/india-nutrient-deficiency-analytics
Live Analytics Dashboard: https://yeshwanthkushi25-glitch.github.io/india-nutrient-deficiency-analytics/